The Augmented Synthetic Historical Control Method

1 Introduction

Causal inference in economic and policy research often compares a treated set of units to a control or donor pool. The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics, second only to difference-in-differences. The idea of SC is straightforward: the untreated potential outcomes of a treated unit are approximated by a weighted average of donor units. Difference-in-differences relies on the assumption of parallel pre-intervention trends [Baker et al., 2025].

In many settings, however, no suitable donor units are available—for example, during global events such as pandemics or financial crises, or when no valid donor units exist. When no valid untreated donor units are available, researchers often turn to the interrupted time series (ITS) design. In its canonical form, ITS models the pre-treatment outcome path as a parametric linear trend and tests for changes in level or slope following an intervention. In fact, even comparative interrupted time series designs have been used by policy analysts [Coopersmith et al., 2022]. Even so, ITS can be fragile in many realistic applications. Its validity depends critically on the assumption that an OLS line fitted to the pre-treatment period provides a reasonable counterfactual projection. In practice, many outcome series feature nonlinear dynamics, seasonality, or structural breaks that violate this assumption. Moreover, ITS typically imposes a homogeneous effect structure, assuming a single jump in level and/or slope that applies equally across all post-treatment periods. These restrictions can lead to biased inference when the true effects are time-varying or when pre-treatment is poor.

To address this shortcoming, Chen et al. [2024] proposed the synthetic historical control method (SHC). Rather than relying on external donors, SHC constructs a donor pool from the treated unit's own pre-treatment history, splitting the pre-treatment period into overlapping segments. The SHC also, unlike ITS, uses kernel smoothing to account for non-linearities in the pre-treatment time series data which ITS does not naturally account for. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, allowing for effect size heterogeneity found in SCM and increasingly DID but not ITS. However, like Abadie et al. [2010, 2015], SHC restricts weights to the simplex. This guarantees interpretability but can be too restrictive when noise or complex dynamics make exact pre-treatment fit unattainable.

This paper introduces the Augmented Synthetic Historical Control method (ASHC), which extends SHC in two key ways. First, ASHC relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This permits negative weights, allowing controlled extrapolation beyond the convex hull of observed segments in exchange for better fit. Second, ASHC adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM [Ben-Michael et al., 2021]. This encourages the estimator to remain close to SHC unless extrapolation is needed. ASHC retains the stabilizing Gram penalty from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings where exact pre-treatment fit is otherwise impossible.

The paper makes three contributions. Methodologically, it introduces the ASHC estimator, which generalizes SHC by permitting limited, penalized extrapolation while maintaining the stabilizing Gram penalty. Theoretically, it derives finite-sample error bounds that decompose ASHC prediction error into four interpretable parts: the SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. It further establishes asymptotic consistency and a convergence rate of $O_p(m^{-H/(2H+3)})$, which approaches the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order H increases. Practically, this means that when the pre-treatment length m is large (e.g. $m = 120$) and H grows, the convergence rate becomes practically close to the parametric $O_p(m^{-1/2})$, so ASHC achieves efficient performance in realistic sample sizes.

While the finite-sample analysis provides guarantees on the accuracy of ASHC for realistic data sizes, asymptotic results are equally important. The finite-sample bounds show how ASHC predictions may differ from SHC depending on regularization strength, operator norm, smoothness bias, and noise. The asymptotic results complement this by demonstrating that, as the amount of pre-treatment data grows, ASHC's prediction error vanishes at a near-parametric rate under mild conditions. Together, these perspectives ensure that ASHC is both practically reliable in finite samples and theoretically sound in large-sample limits.

The discussion begins with a careful review of SHC, its assumptions, and its implementation, before introducing the ASHC framework. We then turn to the theoretical analysis, establishing finite-sample error bounds and asymptotic properties. The empirical performance of ASHC is explored through calibrated simulations, followed by an application that illustrates its usefulness in practice. Proofs are collected in the appendix, and all computations are carried out using Python's mlsynth library (https://mlsynth.readthedocs.io/en/latest/).

For policy analysts, the message is straightforward: ASHC provides a principled way to construct reliable counterfactuals even in environments where no external donor units are available. Furthermore, it allows policy analysts to do this even when methods such as interrupted time series method or the Chen et al. [2024] would not do well.

The remainder of the paper proceeds as follows. Section 1.1 situates our contribution within the existing literature on causal inference in policy evaluation. We then review the SHC method in Section 2.1 before introducing the augmented SHC (ASHC) in Section 2.2. Section 3 presents our theoretical results, including finite-sample error bounds and asymptotic properties. We then turn to an empirical application in Section 4, replicating the analysis of Barcelona's 2015 hotel moratorium from Bel et al. [2021], a setting that allows us to compare ITS, SCM, SHC, and ASHC on equal footing. Finally, Section 5 concludes with implications for both methodological research and policy analysis.

1.1 Literature Review

2 Methodology

2.1 The Synthetic Historical Control Method

Let $\mathcal{T} = \{1, \ldots, T\}$ index time, with pre-treatment period $\mathcal{T}_1 = \{t \in \mathcal{T} : t \leq T_0\}$ and post-treatment period $\mathcal{T}_2 = \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$, and an evaluation window of length m is defined as the final m pre-treatment periods $\mathcal{T}_{\text{eval}} = \{T_0 - m + 1, \ldots, T_0\}$. Let $\mathbf{y} = (y_1, \ldots, y_T)^\top$ denote the observed outcome series. The evaluation subvector is $\mathbf{y}_{\text{eval}} = \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector is $\mathbf{y}_{\text{post}} = \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

For each $t \in \mathcal{T}$, let y_t^1 and y_t^0 denote the potential outcomes under treatment and no treatment, respectively. The observed outcome is

$$y_t = d_t y_t^1 + (1 - d_t) y_t^0, \quad d_t = \mathbf{1}\{t > T_0\}.$$

For $t > T_0$, only y_t^1 is observed. SHC provides an estimate $\widehat{y}_t^0$ for the missing counterfactual, and the treatment effect is estimated as $\widehat{\alpha}_t = y_t - \widehat{y}_t^0, \quad t \in \mathcal{T}_2$, with the average treatment effect on the treated (ATT) defined as $\widehat{\text{ATT}} = n^{-1} \sum_{t \in \mathcal{T}_2} \widehat{\alpha}_t$. The SHC framework is grounded in the following assumptions.

Assumption 2.1 (Error Properties)**.** *The error term satisfies* $\mathbb{E}[\varepsilon_t] = 0$ *and* $\mathbb{E}[\varepsilon_t^2] < \infty$ *for all* $t \in \mathcal{T}$*.*

Assumption 2.2 (Latent Smoothness and Matching)**.** *The latent trend* $\widetilde{y}(t)$ *is* $(H+1)$ *times differentiable for some integer* $H \geq 3$*, with uniformly bounded derivative. Furthermore, there exists* $\mathbf{w}^* \in \Delta^{N-1}$ *such that* $\mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre} \mathbf{w}^*$*, where* Δ^{N-1} *denotes the probability simplex.*

Assumption 2.3 (Feasibility and Stability)**.** *There exists a sparse weight vector* $\mathbf{w} \in \Delta^{N-1}$ *such that* $\mathbf{y}_{eval} \approx \widetilde{\mathbf{Y}}_{pre} \mathbf{w}$ *and the corresponding Gram matrix is positive definite.*

These assumptions ensure that noise is mean-zero and bounded, that the latent component is smooth enough to be locally approximated, and that the evaluation window can be matched stably from the donor pool of pre-treatment segments.

The implementation of SHC begins by smoothing the pre-treatment series $\mathbf{y}_{\text{pre}}$ using local linear regression with a Gaussian kernel, producing a smoothed trajectory $\widetilde{\mathbf{y}}$. From this smoothed series, N overlapping segments of length m are extracted to form the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$. The evaluation vector $\mathbf{y}_{\text{eval}}$ is reconstructed as a convex combination of the donor segments by solving the quadratic program

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2, \qquad \text{subject to } \mathbf{w} \geq \mathbf{0},\ \mathbf{1}^\top \mathbf{w} = 1,$$

where $\mathbf{C}_0$ contains the eigenvectors of the donor Gram matrix associated with eigenvalues below a threshold τ. The additional penalty shrinks weights along low-variance directions, mitigating instability from collinearity. Applying $\mathbf{w}^*$ to the aligned post-treatment donor matrix yields $\widehat{\mathbf{y}}^0_{\text{post}}$, which serves as the SHC estimate of the untreated counterfactual trajectory.

2.2 The Augmented Synthetic Historical Control Method

The augmented synthetic historical control (ASHC) method extends SHC by permitting limited extrapolation beyond the convex hull of historical segments, while stabilizing the solution through a proximity penalty. This modification is motivated by cases where the simplex constraint of SHC is too restrictive to achieve adequate pre-treatment fit. By allowing negative weights under an affine constraint and penalizing deviations from the SHC solution, ASHC balances improved fit with regularization.

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC weight vector. The ASHC estimator solves

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1.$$

As we can see, the first two terms are essentially unchanged. We have a fit term to learn the treated unit's outcomes in the evaluation period and the stability penalty term from Chen et al. [2024]. The third term in the ASHC objective,

$$\lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2,$$

is a proximity penalty. This term discourages the ASHC solution from deviating too far from the original SHC weights. The penalty plays a dual role. Conceptually, it regularizes the affine extension of SHC by shrinking the solution back toward the purely convex combination used in SHC. This ensures that the additional flexibility provided by negative weights is only exercised when it leads to a substantial reduction in pre-treatment error. Mathematically, the penalty stabilizes the optimization problem by making the objective function strongly convex. The tuning parameter $\lambda \geq 0$ controls the degree of shrinkage: when λ is very large, the ASHC solution converges to $\mathbf{w}^{\text{SHC}}$, while when $\lambda = 0$ the penalty vanishes and the estimator reduces to an unpenalized affine extension of SHC. Thus, the proximity penalty creates a continuum between SHC and its affine relaxation, allowing researchers to balance interpretability and stability against the potential gains in fit from extrapolation.

The ASHC counterfactual is obtained as $\widehat{\mathbf{y}}^{0,\text{ASHC}}_{\text{post}} = \widetilde{\mathbf{Y}}_{\text{post}}\, \mathbf{w}^{\text{ASHC}}$, with treatment effects estimated analogously to SHC. In addition to the assumptions underlying SHC, ASHC requires bounds on the operator norms of the donor matrices.

Assumption 2.4 (Operator Norm Bound)**.** *The pre-treatment donor matrix satisfies* $\|\widetilde{\mathbf{Y}}\| \leq \gamma$ *for some constant* $\gamma > 0$.

Assumption 2.5 (Post-Treatment Operator Norm)**.** *There exists* $\gamma_{post} > 0$ *such that* $\|\widetilde{\mathbf{Y}}_{post}\| \leq \gamma_{post}$.

Assumption~2.4 ensures that the smoothed pre-treatment donor matrix does not admit arbitrarily large operator norm growth, which would destabilize the estimation of weights under affine relaxation. Assumption~2.5 provides a corresponding bound for the post-treatment donor matrix, guaranteeing that extrapolation via affine weights cannot yield counterfactual predictions with unbounded variance. Together with the SHC assumptions, these conditions provide stability and control of the ASHC estimator, enabling its theoretical properties to be derived and ensuring its performance in finite samples.

2.3 Bandwidth and Regularization Parameter Selection

The performance of ASHC depends critically on two hyperparameters: the smoothing bandwidth h and the proximity penalty parameter λ. Both are chosen via cross-validation.

First I discuss the bandwidth for the kernel smoothing. Let $\mathbf{y}_{\text{pre}} = (y_1, \ldots, y_{T_0})^\top$ denote the pre-treatment series.
For a candidate bandwidth $h > 0$, define the local linear smoother estimate $\widehat{y}_i^{(-i)}(h)$ as the fitted intercept obtained by regressing $\{y_j : j \neq i\}$ on $\{j - i : j \neq i\}$ with kernel weights

$$K_h(j-i) \;=\; \exp\Big(-\tfrac{1}{2}\left(\tfrac{j-i}{h}\right)^2\Big),$$

normalized so that $\sum_{j \neq i} K_h(j-i) = 1$. Formally, for each $i \in \{1, \ldots, T_0\}$,

$$\widehat{\beta}(h)^{(-i)} = \arg\min_{\beta_0, \beta_1} \sum_{j \neq i} K_h(j-i) \Big(y_j - \beta_0 - \beta_1 (j-i)\Big)^2,$$

and the leave-one-out prediction is

$$\widehat{y}_i^{(-i)}(h) \;=\; \widehat{\beta}_0(h)^{(-i)}.$$

The LOOCV error is

$$\mathrm{CV}(h) \;=\; \frac{1}{T_0} \sum_{i=1}^{T_0} \Big(y_i - \widehat{y}_i^{(-i)}(h)\Big)^2.$$

The optimal bandwidth is then chosen as

$$h^* = \arg\min_{h \in \mathcal{H}} \mathrm{CV}(h),$$

where $\mathcal{H}$ is a prespecified candidate grid. This is part of the normal SHC method.

Next we discuss lambda, exclusive to the ASHC. Recall the ASHC objective

$$Q(\mathbf{w}; \lambda) = \Big\| \mathbf{y}_{\text{train}} - \widetilde{\mathbf{Y}}_{\text{train}} \mathbf{w} \Big\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1.$$

Let the evaluation window $\mathcal{T}_{\text{eval}}$ be split into a training set $\mathcal{T}_{\text{train}}$ and a validation set $\mathcal{T}_{\text{val}}$, with corresponding subvectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}$ and donor matrices $\widetilde{\mathbf{Y}}_{\text{train}}, \widetilde{\mathbf{Y}}_{\text{val}}$.

For each $\lambda > 0$,

$$\widehat{\mathbf{w}}(\lambda) = \arg\min_{\mathbf{w} \in \mathbb{R}^N} Q(\mathbf{w}; \lambda),$$

and the validation error is

$$\mathrm{Val}(\lambda) = \frac{1}{|\mathcal{T}_{\text{val}}|} \left\| \mathbf{y}_{\text{val}} - \widetilde{\mathbf{Y}}_{\text{val}} \, \widehat{\mathbf{w}}(\lambda) \right\|_2^2.$$

The optimal penalty parameter is selected as

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathrm{Val}(\lambda),$$

where Λ is a candidate grid of λ values (e.g. logarithmically spaced).

Together, h^* and λ^* balance the dual bias–variance tradeoffs of ASHC: smoothing bias versus noise, and extrapolation bias versus instability.

3 Theoretical Findings

We provide finite sample and asymptotic analysis of the ASHC estimator.

Lemma 3.1 (Deviation from SHC)**.** *Let* $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$*. Then*

$$\|\Delta\| \ \leq \ \frac{\gamma}{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}\, \epsilon_{SHC}.$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ *and* $\|\mathbf{C}_0\|^2$ *is the operator norm of* $\mathbf{C}_0^\top \mathbf{C}_0$*.*

Lemma~3.1 establishes that the deviation between ASHC and SHC weights, $\|\Delta\| = \|\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}\|$, is bounded by $\frac{\gamma}{2\lambda+2\varsigma\|\mathbf{C}_0\|^2}\epsilon_{\mathrm{SHC}}$, where ϵ_{SHC} is the SHC in-sample fit error.

Corollary 3.1. *The primary driver of the deviation* $\|\Delta\|$ *between the ASHC and SHC weights is the SHC residual error* ϵ_{SHC}*. The terms* λ *and* $\lambda_{\min}(M)$ *appear in the denominator and therefore act only as scaling factors: stronger regularization (*λ *large) or greater curvature of the Hessian (*$\lambda_{\min}(M)$ *large) shrink the deviation, but cannot generate deviation on their own. In contrast,* ϵ_{SHC} *enters multiplicatively in the numerator. Thus, if the SHC fit is already tight (so that* ϵ_{SHC} *is small), the deviation* $\|\Delta\|$ *is necessarily small. Conversely, if SHC exhibits poor fit, then* ϵ_{SHC} *is large, and ASHC permits correspondingly larger deviations, though still bounded by the regularization terms. In this way, ASHC improves stability without moving far from the baseline SHC solution unless the SHC fit leaves room for meaningful improvement.*

To analyze the ASHC estimator relative to the SHC estimator, we aim to bound how much the ASHC predictions can deviate given a change in the weight vector. We do this using the operator norm of the donor matrix $\widetilde{\mathbf{Y}}$, which measures the largest possible amplification of weight deviations into prediction deviations.

Proposition 3.1. *Under assumptions (A1)–(A5), the ASHC prediction error is bounded by*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \ \leq \ \epsilon_{SHC}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

Proposition~3.1 bounds the ASHC pre-treatment prediction error by $\epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda+2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta$, incorporating the SHC error, smoothness bias, and noise. For post-treatment predictions, Lemma~**??** bounds the approximation error by $b_{\epsilon,\mathrm{post}}(H,k) + \delta_{\mathrm{post}}$, reflecting smoothness and noise contributions. Analogous bounds may be made for the post-treatment regime, too.

Proposition 3.2 (Out-of-sample prediction and ATT error bound)**.** *Under Assumptions (A1)–(A5) and Assumption 2.5, the ASHC estimator's post-treatment prediction error satisfies*

$$\left\|\mathbf{y}_{post} - \mathbf{Y}_{post}\mathbf{w}^{ASHC}\right\| \leq b_{\epsilon,post}(H,k) + \delta_{post} + \gamma_{post}\left(\eta + \frac{\epsilon_{SHC}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right),$$

where η *bounds the difference* $\|\mathbf{w}^* - \mathbf{w}^{SHC}\|$ *and* ϵ_{SHC} *is the SHC in-sample fit error.*

Moreover, the error in the average treatment effect on the treated (ATT) estimate is bounded by

$$\left|\widehat{ATT} - ATT\right| \leq \frac{1}{\sqrt{n}}\left[b_{\epsilon,post}(H,k) + \delta_{post} + \gamma_{post}\left(\eta + \frac{\epsilon_{SHC}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right)\right].$$

Proposition~3.2 extends this to the post-treatment error, bounded by $b_{\epsilon,\mathrm{post}}(H,k)+\delta_{\mathrm{post}}+\gamma_{\mathrm{post}}\left(\eta + \frac{\epsilon_{\mathrm{SHC}}}{2\lambda+2\varsigma\|\mathbf{C}_0\|^2}\right)$, where η bounds the difference between oracle and SHC weights (Lemma~**??**). The Average Treatment Effect on the Treated (ATT) error is similarly bounded, scaled by $1/\sqrt{n}$.

Now we return to asymptotics.

Theorem 3.1. *Under (A1)–(A5), as* $m \to \infty$ *and* $H \to \infty$*, for fixed* $k > 0$*,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \longrightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}} w^{ASHC}\| \xrightarrow{p} 0.$$

Theorem~3.1 proves that as the evaluation window $m \to \infty$ and Taylor expansion order $H \to \infty$, the ASHC prediction error converges to the noise bound δ. Under sub-Gaussian noise, the error converges in probability to zero, ensuring consistency. These results confirm that ASHC maintains stability relative to SHC while improving fit when SHC residuals are large, with robust error bounds and asymptotic consistency under standard assumptions.

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4 Empirical Application

We replicate the setting of Bel et al. [2021], who study the effects of Barcelona's 2015 hotel moratorium using data on hotel prices from 2010 to 2017. This application provides a compelling case to benchmark ITS, SCM, SHC, and ASHC against one another.

5 Conclusion

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